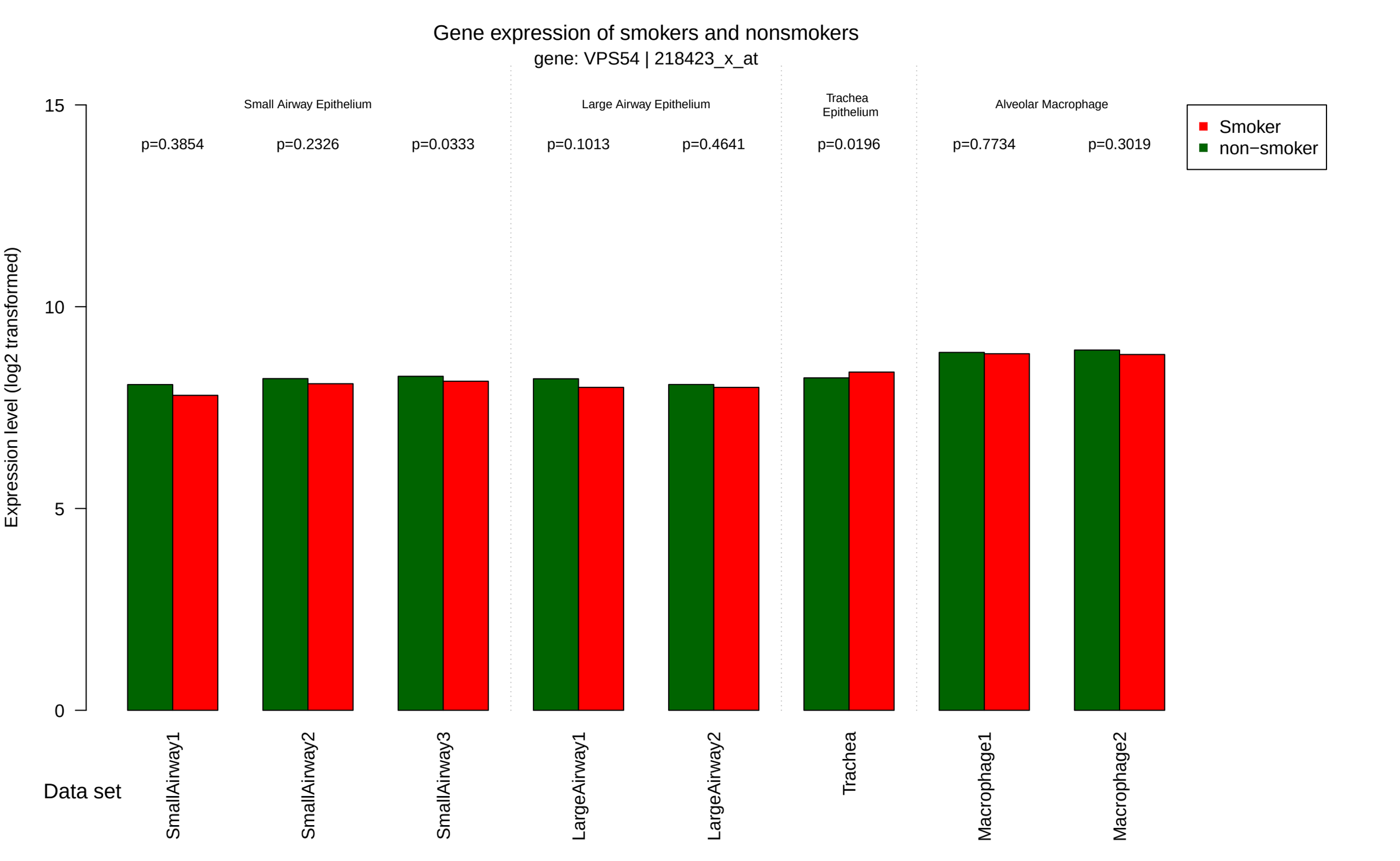
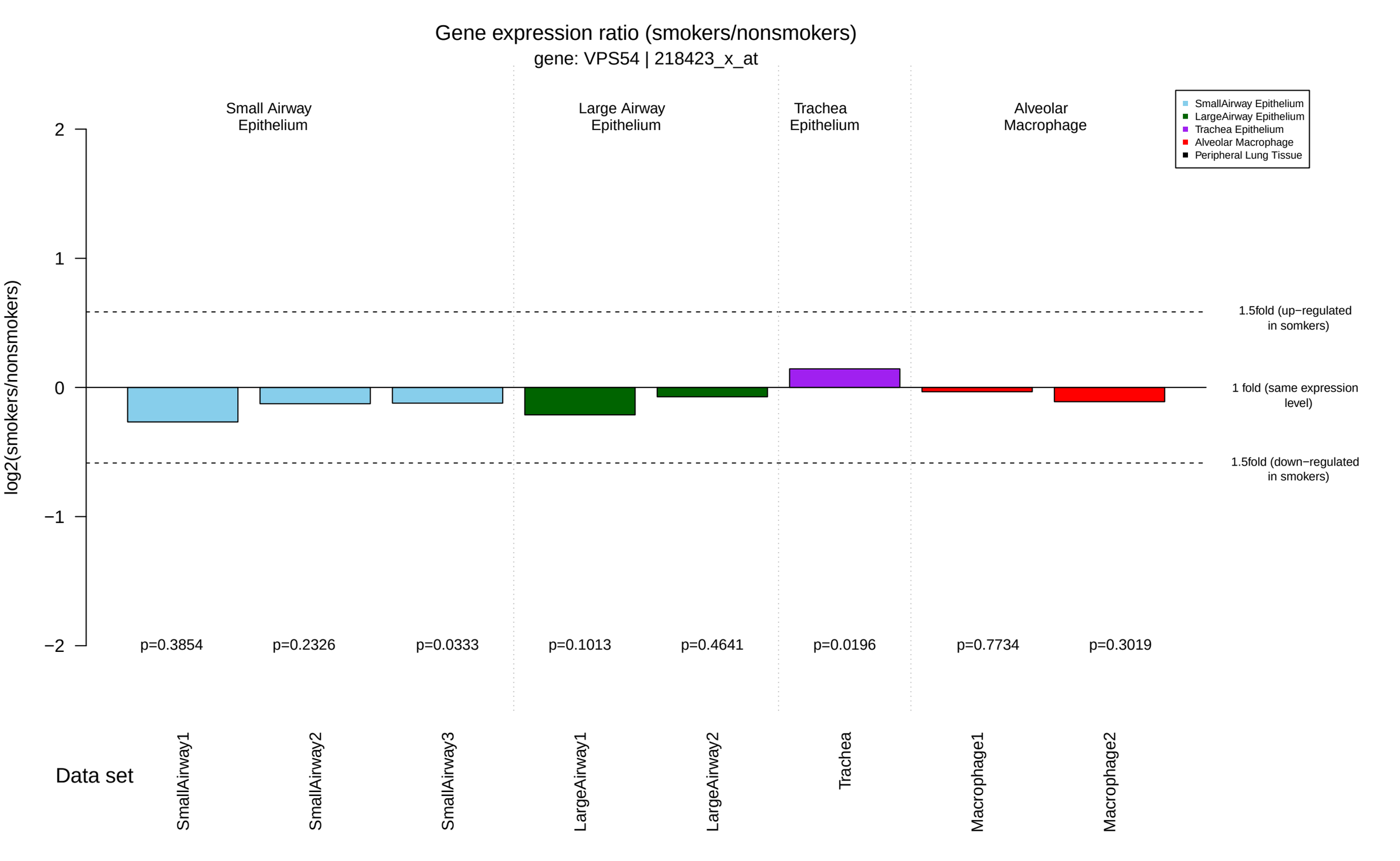
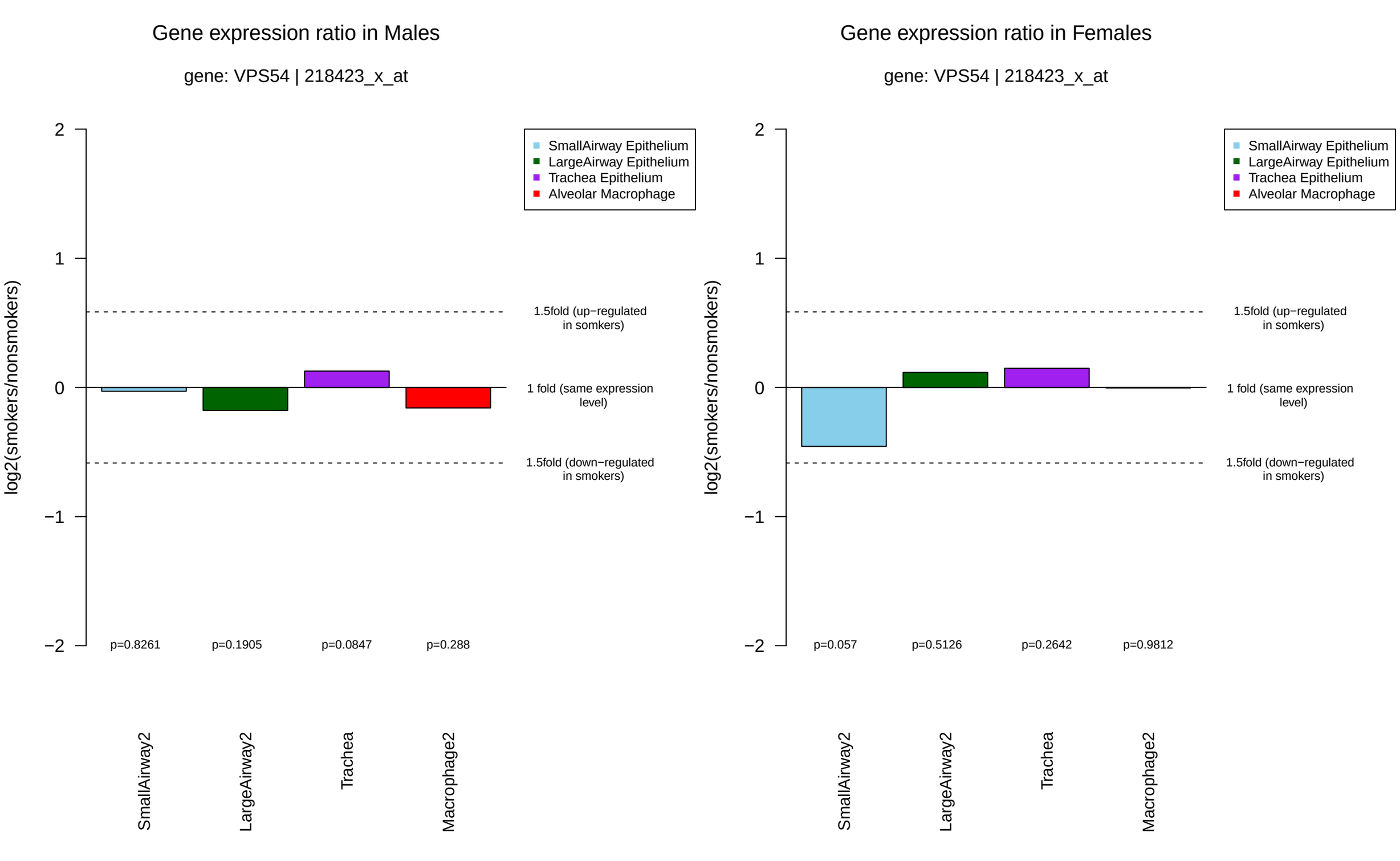
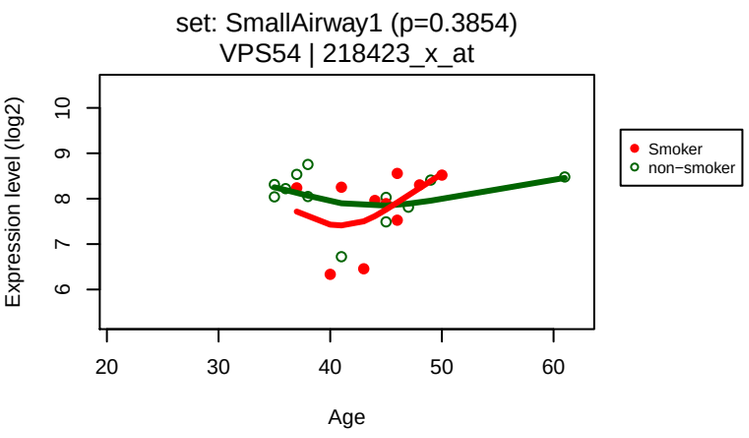
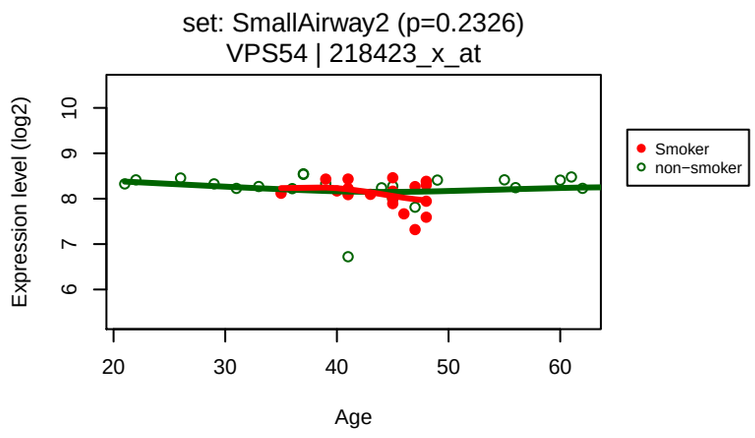
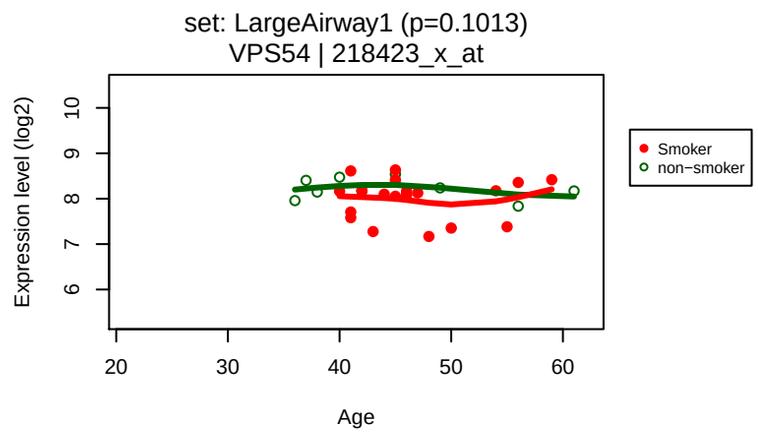

Gene expression of smokers and nonsmokers
gene: VPS54 | 218423_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3854
p=0.2326
p=0.0333
p=0.1013
p=0.4641
p=0.0196
p=0.7734
p=0.3019
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

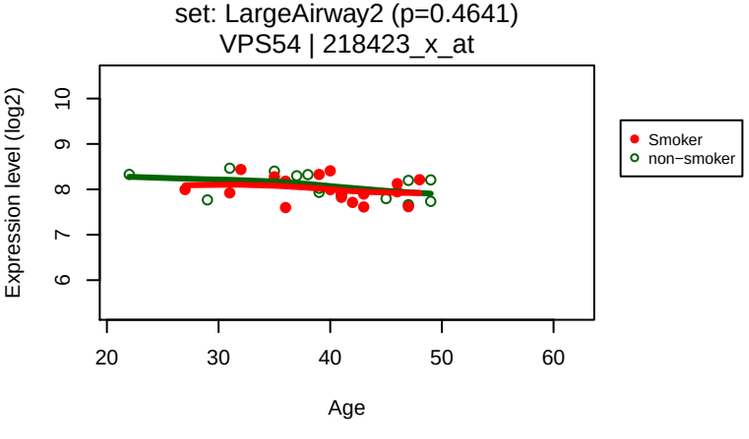

Gene expression ratio (smokers/nonsmokers)
gene: VPS54 | 218423_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3854
p=0.2326
p=0.0333
p=0.1013
p=0.4641
p=0.0196
p=0.7734
p=0.3019
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

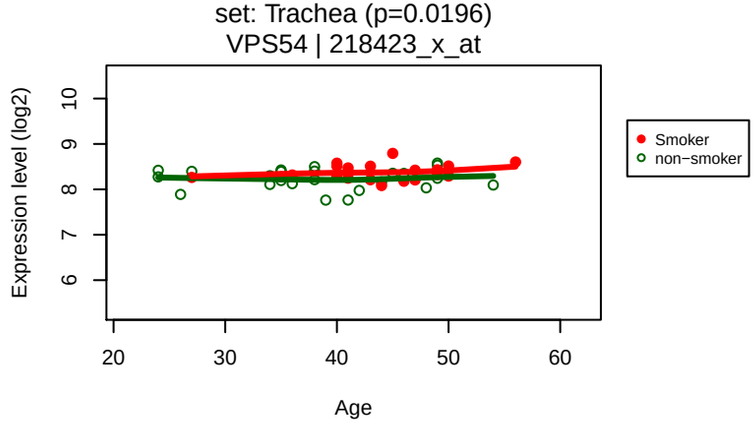

Gene expression ratio in Males
gene: VPS54 | 218423_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8261
p=0.1905
p=0.0847
p=0.288
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VPS54 | 218423_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.057
p=0.5126
p=0.2642
p=0.9812
SmallAirway2
LargeAirway2
Trachea
Macrophage2

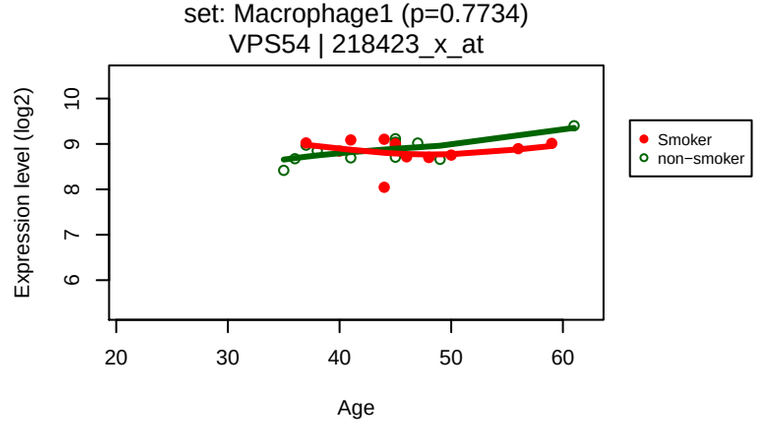

set: SmallAirway1 (p=0.3854)
VPS54 | 218423_x_at
Expression level (log2)
Age
Smoker
non−smoker

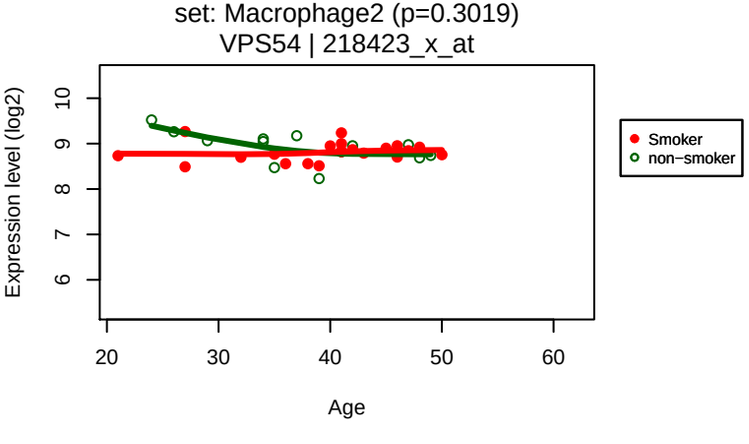

set: SmallAirway2 (p=0.2326)
VPS54 | 218423_x_at
Expression level (log2)
Age
Smoker
non−smoker

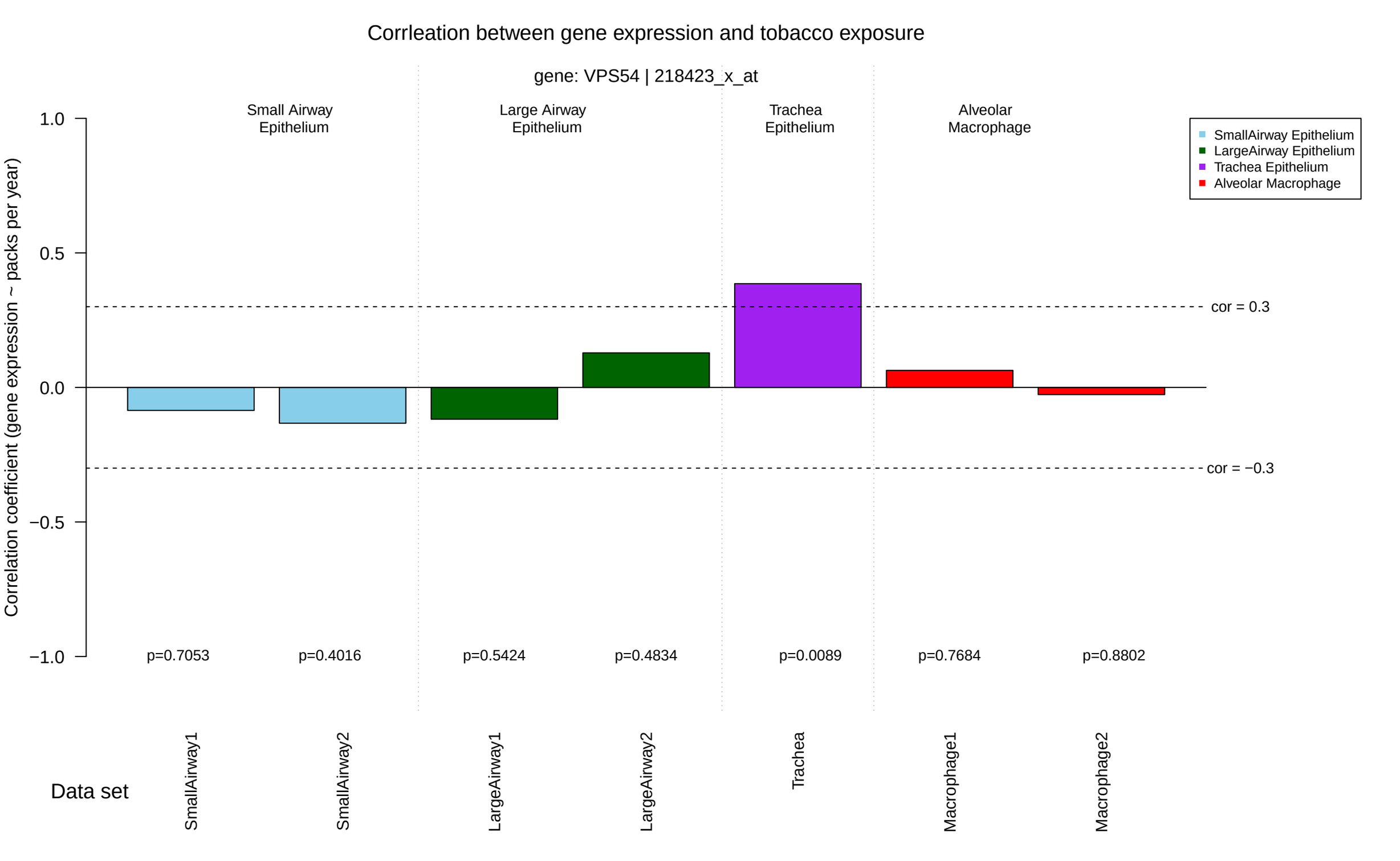

set: LargeAirway1 (p=0.1013)
VPS54 | 218423_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4641)
VPS54 | 218423_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0196)
VPS54 | 218423_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7734)
VPS54 | 218423_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3019)
VPS54 | 218423_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VPS54 | 218423_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7053
p=0.4016
p=0.5424
p=0.4834
p=0.0089
p=0.7684
p=0.8802
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2